TYLCV-IS76 Positive-Control Validation

Result

The SeqScape recombination wrapper recovered the known TYLCV-IS76 resistance-breaking recombinant signal. In the strict screen (`p <= 0.05`), the consensus event resolves:

- recombinant: `IS76_LN812978`
- backbone parent: `TYLCV_IL_AM409201`
- donor parent: `TYLCSV_NC_003828`
- breakpoint interval: alignment columns `1382–1492`
- strict supporting method: `geneconv`
- role support: `628` informative sites

The audit run retains RDP's score-like output and merges it with GENECONV over the same tract envelope (`1343–1492`). This is reported separately because the RDP output column is not a valid probability in this OpenRDP run (`36.36`, greater than 1), while GENECONV gives the strict significant call.

Input

Input alignment:

`/Users/gerritkoorsen/ciderseq-mono/cideseq-mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/is76_control_alignment.fasta`

Positive-control metadata:

`/Users/gerritkoorsen/ciderseq-mono/cideseq-mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/is76_positive_control.json`

role	id	description
known recombinant	IS76_LN812978	TYLCV-IS76 isolate G8 (Morocco)
backbone parent	TYLCV_IL_AM409201	TYLCV-IL isolate RE4 (Reunion, 2004)
donor parent	TYLCSV_NC_003828	TYLCSV reference

The metadata file records the published TYLCV-IS76 tract as approximately 76 nt. The local control annotation places the recovered signal in the intergenic region immediately downstream of the conserved origin motif, with a reconstructed tract range of 57-103 nt.

Recombination Calls

run	recombinant	backbone_parent	donor_parent	methods	breakpoint_columns	min_pvalue	role_support
strict p<=0.05	IS76_LN812978	TYLCV_IL_AM409201	TYLCSV_NC_003828	geneconv	1382-1492	0.0	628
RDP audit p<=1.0	IS76_LN812978	TYLCV_IL_AM409201	TYLCSV_NC_003828	geneconv,rdp	1343-1492	0.0	628

All OpenRDP methods completed in the strict run:

method	strict_status
rdp	ok
geneconv	ok
maxchi	ok
chimaera	ok
threeseq	ok
bootscan	ok
siscan	ok

Genome-Derived Pipeline Check

To test the practical workflow from comparator genomes, the three positive-control records were extracted from:

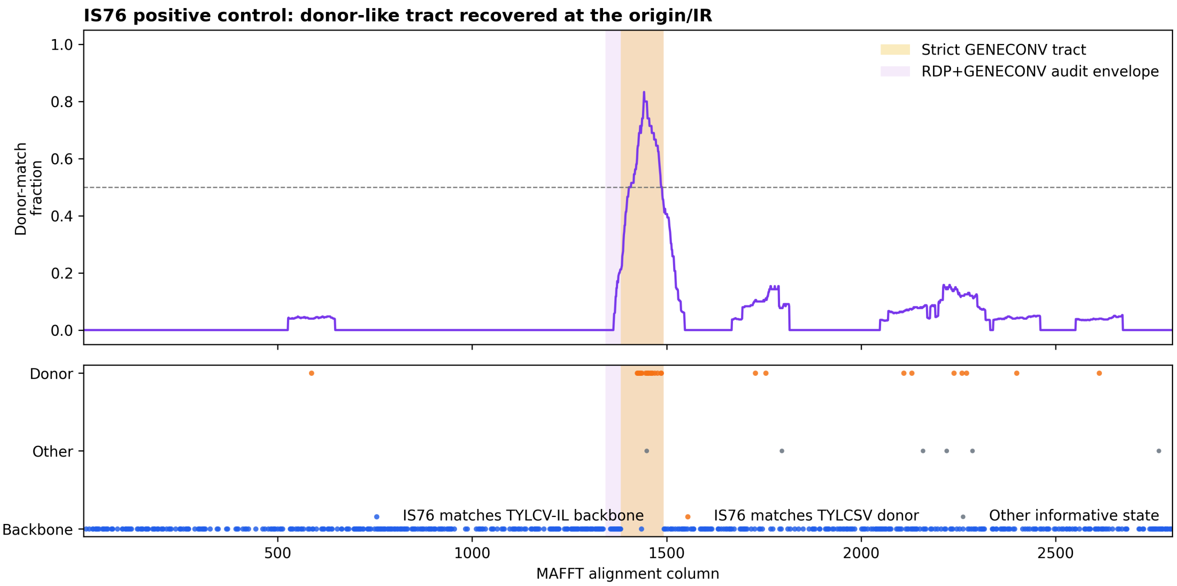
`/Users/gerritkoorsen/ciderseq-mono/cideseq-mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/comparator_panel.fasta`

Origin-centered windows were cut around the conserved `TAATATTAC` motif, aligned with MAFFT, and passed through the same SeqScape recombination wrapper.

input	records	consensus	recombinant	backbone_parent	donor_parent	methods	breakpoint_columns	min_pvalue	role_support	interpretation
400 nt origin window from comparator genomes	3	detected	not resolved	not resolved	not resolved	bootscan,geneconv	100-304	0.0	0	event detected, but the window is too short for flank-based role resolution
1000 nt origin window from comparator genomes	3	detected	LN812978	AM409201	NC_003828	geneconv	494-604	0.0	236	event and expected parent roles recovered

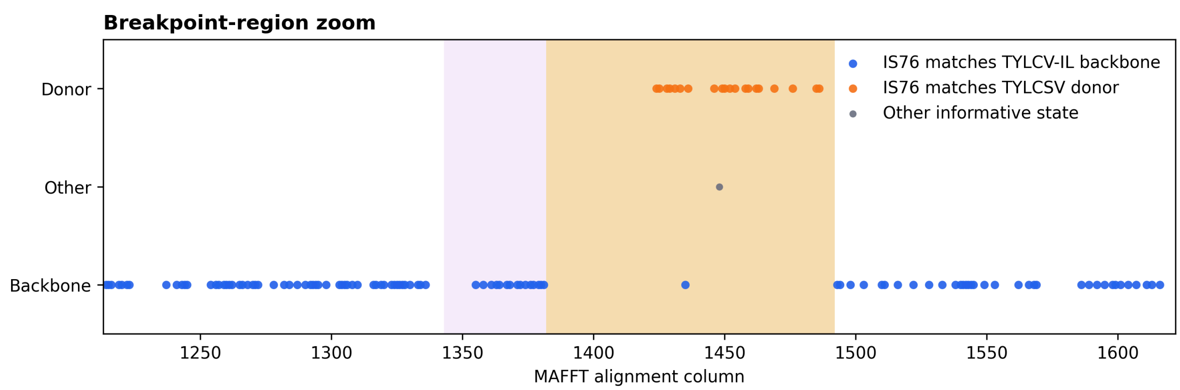
The 400 nt window is sensitive enough to detect the event, but its significant intervals consume most of the window and leave no informative flank support for parent-role polarisation. The 1000 nt origin window resolves the expected roles directly from the extracted genomes: `LN812978` as recombinant, `AM409201` as backbone parent, and `NC_003828` as donor parent.

Figures



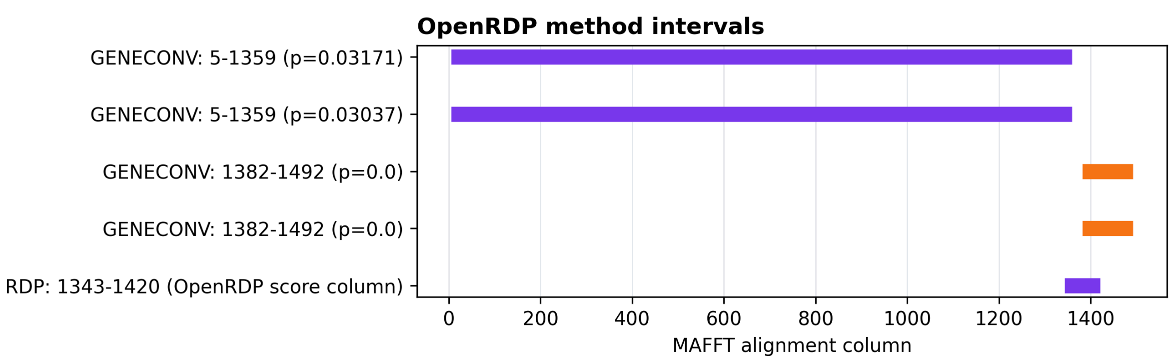
Parent-state scan

Figure 1. Informative-site scan across the aligned TYLCV-IS76 positive control. Blue sites are columns where IS76 matches the TYLCV-IL backbone parent; orange sites are columns where IS76 matches the TYLCSV donor parent. The strict GENECONV tract is shaded yellow. The RDP+GENECONV audit envelope is shaded purple.



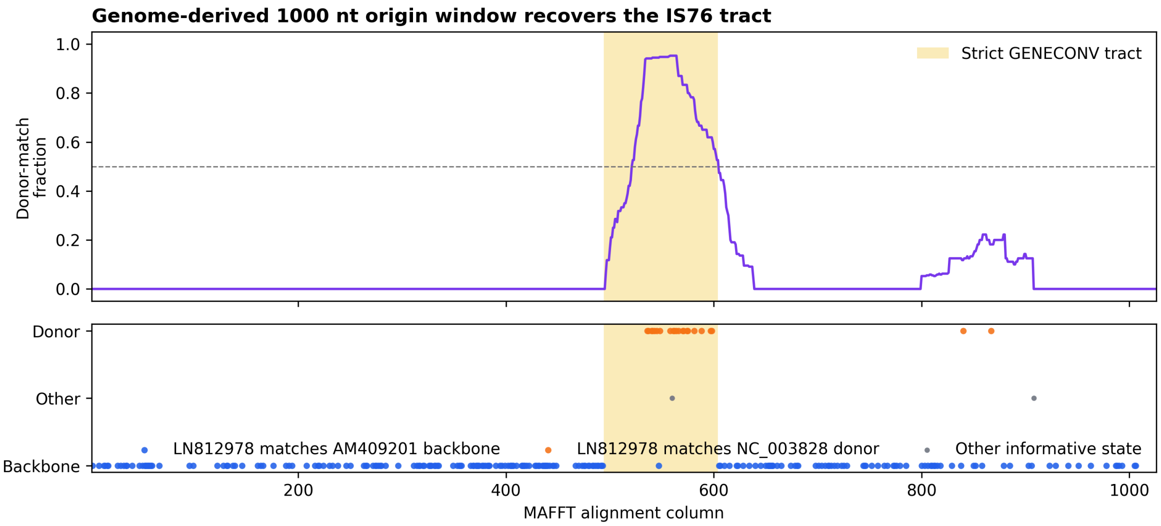
Breakpoint zoom

Figure 2. Zoom around the recovered breakpoint region. The donor-like informative sites concentrate inside the OpenRDP/GENECONV interval, while the flanking sequence returns to the TYLCV-IL backbone state.



Method intervals

Figure 3. OpenRDP method intervals in the positive-control run. GENECONV gives the strict significant interval; RDP reports an overlapping interval but its numeric field is score-like rather than a p-value in this run.



Genome-derived origin-window scan

Figure 4. Informative-site scan for the 1000 nt origin-centered window extracted from the comparator genome FASTA. The donor-like sites concentrate in the strict GENECONV interval, while flanking sites support the TYLCV-IL backbone parent.

Interpretation

This positive-control test passes. The pipeline detects the known resistance-breaking recombination pattern and assigns the expected roles: IS76 as recombinant, TYLCV-IL as the backbone parent, and TYLCSV as the donor-side parent.

The strict result is intentionally conservative: it promotes the GENECONV-supported event under `p <= 0.05`. The RDP audit supports the same region geometrically, but is not counted as strict statistical support because OpenRDP emits a value outside the probability range for that method in this control.

For full-genome inputs, this validation supports using an origin-centered window with enough flanking sequence for role assignment. A 1000 nt window worked for this positive control; a 400 nt window detected recombination but did not retain enough flanking signal to resolve the parent roles.

Informative-site counts from the alignment:

- donor-matching informative sites: 30
- backbone-matching informative sites: 598
- other informative states: 6
- total informative sites: 634

The `role_support` value is the donor-matching plus backbone-matching count (`628`); the six other informative states are retained in the figure but are not counted as role support.

Commands

Strict run:

```
PATH="/private/tmp/openrdp-validation-venv/bin:$PATH" PYTHONPATH=src \
python -m seqscape.cli recombination \
  --alignment-fasta "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/is76_control_alignment.fasta" \
  --methods rdp,geneconv,maxchi,chimaera,threeseq,bootscan,siscan \
  --min-methods 1 \
  --pvalue 0.05 \
  --outdir "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/strict_p005"
```

RDP audit run:

```
PATH="/private/tmp/openrdp-validation-venv/bin:$PATH" PYTHONPATH=src \
python -m seqscape.cli recombination \
  --alignment-fasta "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/is76_control_alignment.fasta" \
  --methods rdp,geneconv,maxchi,chimaera,threeseq,bootscan,siscan \
  --min-methods 1 \
  --pvalue 1.0 \
  --outdir "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/rdp_audit_p1"
```

Genome-derived 1000 nt origin-window run:

```
python scripts/extract_origin_windows.py \
  --fasta "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/runs/tocsv_260203_full_aws_20260612/full_run/comparators/comparator_panel.fasta" \
  --ids "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/inputs/is76_ids.txt" \
  --window-size 1000 \
  --out-fasta "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/is76_origin_window
        _1000nt.fasta" \
  --manifest-tsv "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/is76_origin_window
        _1000nt_manifest.tsv"

mafft --auto --thread 8 \
  "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/is76_origin_window
        _1000nt.fasta" \
  > "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/is76_origin_window
        _1000nt_aligned.fasta"

PATH="/private/tmp/openrdp-validation-venv/bin:$PATH" PYTHONPATH=src \
python -m seqscape.cli recombination \
  --alignment-fasta "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/is76_origin_window
        _1000nt_aligned.fasta" \
  --methods rdp,geneconv,maxchi,chimaera,threeseq,bootscan,siscan \
  --min-methods 1 \
  --pvalue 0.05 \
  --outdir "/Users/gerritkoorsen/ciderseq-mono/cideseq-
        mono/seqscape/runs/tylcv_is76_pipeline_validation_20260814/from_genomes_1000nt/recombination_stri
        ct_p005"
```